STM32F429I Discovery

Overview

The STM32F429I-DISC1 Discovery kit features an ARM Cortex-M4 based STM32F429ZI MCU with a wide range of connectivity support and configurations. Here are some highlights of the STM32F429I-DISC1 board:

- STM32 microcontroller in LQFP144 package
- Extension header for all LQFP144 I/Os for quick connection to prototyping board and easy probing
- On-board ST-LINK/V2-B debugger/programmer with SWD connector
- Flexible board power supply:
 - ST-LINK/V2-1 USB connector
 - User USB FS connector
 - VIN from Arduino* compatible connectors
- Two push-buttons: USER and RESET
- USB OTG FS with micro-AB connector
- 2.4-inch QVGA LCD with MIPI DSI interface and capacitive touch screen
- 64Mbit SDRAM
- L3GD20, ST-MEMS motion sensor 3-axis digital output gyroscope
- Six LEDs
 - LD1 (red/green) for USB communication
 - LD2 (red) for 3.3 V power-on
 - Two user LEDs: LD3 (green), LD4 (red)
 - Two USB OTG LEDs: LD5 (green) VBUS and LD6 (red) OC (over-current)

Board Overview



STM32F429I Discovery

Name: stm32f429i_disc1
Vendor: STMicroelectronics
Architecture: arm
SoC: stm32f429xx

Browse board sources

More information about the board can be found at the STM32F429I-DISC1 website.

Hardware

The STM32F429I-DISC1 Discovery kit provides the following hardware components:

- STM32F429ZIT6 in LQFP144 package
- ARM® 32-bit Cortex® -M4 CPU with FPU
- 180 MHz max CPU frequency
- VDD from 1.8 V to 3.6 V
- 2 MB Flash
- 256+4 KB SRAM including 64-Kbyte of core coupled memory
- GPIO with external interrupt capability
- 3x12-bit ADC with 24 channels
- 2x12-bit D/A converters
- RTC
- Advanced-control Timer
- General Purpose Timers (17)
- Watchdog Timers (2)
- USART/UART (4/4)
- I2C (3)
- SPI (6)
- SDIO
- 2xCAN
- USB 2.0 OTG FS with on-chip PHY
- USB 2.0 OTG HS/FS with dedicated DMA, on-chip full-speed PHY and ULPI
- 10/100 Ethernet MAC with dedicated DMA
- 8- to 14-bit parallel camera
- CRC calculation unit
- True random number generator
- DMA Controller

More information about STM32F429ZI can be found here:

- STM32F429ZI on www.st.com
- STM32F429 Reference Manual

Supported Features

The `stm32f429i_disc1` board supports the hardware features listed below.

on-chip / on-board — Feature integrated in the SoC / present on the board.
×2 / ×2 — Number of instances that are enabled / disabled. Click on the label to see the first instance of this feature in the board/SoC DTS files.
vnd,foo — Compatible string for the Devicetree binding matching the feature. Click on the link to view the binding documentation.

stm32f429i_disc1 / stm32f429xx

TYPE	LOCATION	DESCRIPTION	COMPATIBLE
CPU	on-chip	ARM Cortex-M4F CPU ×1	arm,cortex-m4f
ADC	on-chip	STM32F4 ADC ×1	st,stm32f4-adc
	on-chip	STM32 ADC ×2	st,stm32-adc
CAN	on-chip	STM32 CAN controller ×2	st,stm32-bxcan
Clock control	on-chip	STM32F4 RCC (Reset and Clock controller) ×1	st,stm32f4-rcc
	on-chip	STM32 HSE Clock ×1	st,stm32-hse-clock
	on-chip	Generic fixed-rate clock provider ×1 ×2	fixed-clock
	on-chip	STM32F4 Main PLL ×1	st,stm32f4-pll-clock
	on-chip	STM32F4 PLL I2S ×1	st,stm32f4-plli2s-clock
	on-chip	PLLSAI node binding for STM32F4 and STM32F7 device ×1	st,stm32fx-pllsai-clock
	on-chip	STM32 Microcontroller Clock Output (MCO) ×2	st,stm32-clock-mco
Counter	on-chip	STM32 counters ×14	st,stm32-counter
DAC	on-chip	STM32 family DAC ×1	st,stm32-dac
Display	on-chip	STM32 LCD-TFT display controller ×1	st,stm32-ltdc
	on-board	Ilitek ILI9341 320x240 display controller ×1	ilitek,ili9341
DMA	on-chip	STM32 DMA controller (V1) ×2	st,stm32-dma-v1
Ethernet	on-chip	STM32 Ethernet Controller ×1	st,stm32-ethernet-control…
	on-chip	ST STM32 Ethernet MAC, a child node of the Ethernet controller ×1	st,stm32-ethernet
Flash controller	on-chip	STM32 Family flash controller ×1	st,stm32-flash-controller
GPIO & Headers	on-chip	STM32 GPIO Controller ×11	st,stm32-gpio
I2C	on-chip	STM32 I2C V1 controller ×3	st,stm32-i2c-v1
I2S	on-chip	STM32 I2S controller ×2	st,stm32-i2s
	on-chip	STM32 SAI controller ×2	st,stm32-sai
Input	on-board	STMPE811 I2C touchscreen controller ×1	st,stmpe811
	on-board	Group of GPIO-bound input keys ×1	gpio-keys
Interrupt controller	on-chip	ARMv7-M NVIC (Nested Vectored Interrupt Controller) ×1	arm,v7m-nvic
	on-chip	STM32 External Interrupt Controller ×1	st,stm32-exti
LED	on-board	Group of GPIO-controlled LEDs ×1	gpio-leds
MDIO	on-chip	STM32 MDIO Controller ×1	st,stm32-mdio
Memory controller	on-chip	STM32 Battery Backed RAM ×1	st,stm32-bbram
	on-chip	STM32 Flexible Memory Controller (FMC) ×1	st,stm32-fmc
	on-chip	STM32 Flexible Memory Controller (SDRAM controller) ×1	st,stm32-fmc-sdram
MMC	on-chip	STM32 SDMMC Disk Access ×1	st,stm32-sdmmc
MTD	on-chip	STM32F4 flash memory ×1	st,stm32f4-nv-flash
PHY	on-chip	This binding is to be used by all the usb transceivers which are built-in with USB IP ×2	usb-nop-xceiv
Pin control	on-chip	STM32 Pin controller ×1	st,stm32-pinctrl
Power management	on-chip	STM32 power controller ×1	st,stm32-pwr
PWM	on-chip	STM32 PWM ×12	st,stm32-pwm
Reset controller	on-chip	STM32 Reset and Clock Control (RCC) Controller ×1	st,stm32-rcc-rctl
RNG	on-chip	STM32 Random Number Generator ×1	st,stm32-rng
RTC	on-chip	STM32 RTC ×1	st,stm32-rtc
Sensors	on-chip	STM32 quadrature decoder ×6	st,stm32-qdec
	on-chip	STM32 family TEMP node for production calibrated sensors with two calibration temperatures ×1	st,stm32-temp-cal
	on-chip	STM32 VREF+ ×1	st,stm32-vref
	on-chip	STM32 VBAT ×1	st,stm32-vbat
Serial controller	on-chip	STM32 USART ×1 ×3	st,stm32-usart
	on-chip	STM32 UART ×4	st,stm32-uart
SMbus	on-chip	STM32 SMBus controller ×3	st,stm32-smbus
SPI	on-chip	STM32 SPI controller ×1 ×5	st,stm32-spi
Timer	on-chip	ARMv7-M System Tick ×1	arm,armv7m-systick
	on-chip	STM32 timers ×14	st,stm32-timers
USB	on-chip	STM32 OTGFS controller ×1	st,stm32-otgfs
	on-chip	STM32 OTGHS controller ×1	st,stm32-otghs
Watchdog	on-chip	STM32 watchdog ×1	st,stm32-watchdog
	on-chip	STM32 system window watchdog ×1	st,stm32-window-watchdog

Pin Mapping

The STM32F429I-DISC1 Discovery kit has 8 GPIO controllers. These controllers are responsible for pin muxing, input/output, pull-up, etc.

For more details please refer to STM32F429I-DISC1 board User Manual.

Default Zephyr Peripheral Mapping:

- UART_1_TX : PA9
- UART_1_RX : PA10
- USER_PB : PA0
- LD3 : PG13
- LD4 : PG12
- I2C_1_SCL : PB8
- I2C_1_SDA : PB9
- I2C_2_SCL : PB10
- I2C_2_SDA : PB11
- I2C_3_SCL : PA8
- I2C_3_SDA : PC9
- SPI_5_CS : PF6
- SPI_5_SCK : PF7
- SPI_5_MISO : PF8
- SPI_5_MOSI : PF9
- OTG_HS_ID : PB12
- OTG_HS_DM : PB14
- OTG_HS_DP : PB15

System Clock

The STM32F429I-DISC1 System Clock could be driven by an internal or external oscillator, as well as by the main PLL clock. By default the system clock is driven by the PLL clock at 168MHz, driven by an 8MHz high speed external clock.

Serial Port

The STM32F429I-DISC1 Discovery kit has up to 8 UARTs. The Zephyr console output is assigned to UART1. The default communication settings are 115200 8N1.

USB Port

The STM32F429I-DISC1 Discovery kit has a USB FS capable Micro-B port. It is connected to the on-chip OTG_HS peripheral, but operates in FS mode only since no HS PHY is present. The board supports device and host OTG operation, but only device mode has been tested with Zephyr at this time.

Programming and Debugging

The `stm32f429i_disc1` board supports the runners and associated west commands listed below.

	flash	debug	rtt	attach	debugserver
jlink	✅	✅	✅	✅	✅
openocd	✅	✅ (default)	✅	✅	✅
pyocd	✅	✅	✅	✅	✅
stm32cubeprogrammer	✅ (default)				

The STM32F429I-DISC1 Discovery kit includes a ST-LINK/V2-B embedded debug tool interface. Applications for the `stm32f429i_disc1` board configuration can be built and flashed in the usual way (see Building an Application and Run an Application for more details).

Flashing

The board is configured to be flashed using west STM32CubeProgrammer runner, so its installation is required.

Alternatively, OpenOCD, JLink, or pyOCD can also be used to flash the board using the `--runner` (or `-r`) option:

```
$ west flash --runner openocd
$ west flash --runner jlink
$ west flash --runner pyocd
```

First, connect the STM32F429I-DISC1 Discovery kit to your host computer using the USB port to prepare it for flashing. Then build and flash your application.

Here is an example for the Hello World application.

```
# From the root of the zephyr repository
west build -b stm32f429i_disc1 samples/hello_world
west flash
```

Run a serial host program to connect with your board:

```
$ minicom -D /dev/ttyACM0
```

Then, press the RESET button (The black one), you should see the following message:

```
Hello World! arm
```

Debugging

You can debug an application in the usual way. Here is an example for the Hello World application.

```
# From the root of the zephyr repository
west build -b stm32f429i_disc1 samples/hello_world
west debug
```

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